JAWAHARLAL NEHRU PORT AUTHORITY

INTERNSHIP AT JNPA REPORT BY

Mr. Suyog Sudhakar Gorule

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CERTIFICATE

This is to certify that Mr. Suyog Sudhakar Gorule from Fr. Conceicao Rodrigues Institute of Technology(Autonomous), Vashi has completed internship as Web Developer at JAWAHARLAL NEHRU PORT AUTHORITY, SHEVA , Navi Mumbai Maharashtra.

This report is the record of work carried out by them during the period from 9th June 2025 to 8th July 2025.

Signature of Guide

Signature of Guide

Date:

ACKNOWLEDGEMENT

I would like to sincerely thank the Fr. Conceicao Rodrigues Institute of Technology (Autonomous), Vashi for providing me the opportunity to undertake my training at Jawaharlal Nehru Port Authority.

My heartfelt gratitude goes to Shri Suhas Kamtikar and Mr. Shishir Bansode, Sr. Manager (Personnel), for offering me valuable opportunities and insightful guidance throughout the course of my training.

I am especially thankful to Shri Sachin Shinde and Shri Prakash Ingale for their dedicated support and expert advice, which proved immensely helpful in enriching both my theoretical understanding and practical knowledge. Their involvement in key decisions and continuous assistance greatly eased my learning experience.

Lastly, I extend my appreciation to all the employees who took the time to share their knowledge and support me during this internship. I am gratefully to acknowledge their contributions at this moment.

ACRONYMS

JNPA	Jawaharlal Nehru Port Authority
ADM	Administration Building
MS	Management Services
CTCC	Container Terminal Computer Centre
SAP	System Application Product
RDBMS	Relational Database Management System
RMQC	Rail Mounted Quary Crane
TEU	Twenty-foot equivalent unit
CRS	Contract Reminder System

ABSTRACT

In today's rapidly advancing industrial and technological landscape, understanding the operational intricacies of large-scale organizations is essential for bridging the gap between theoretical learning and real-world application. This internship report is a comprehensive account of the professional experience gained during my tenure at Jawaharlal Nehru Port Authority (JNPA), India's largest and most pivotal container port. The internship was designed to provide exposure to the organizational structure, operational workflows, technical systems, and digital transformation initiatives of JNPA.

Throughout the internship, I was exposed to various departments including IT, operations, infrastructure, and administration. A significant portion of the training focused on the use of enterprise software tools like SAP and Oracle, which are integral to the organization's data management, asset tracking, finance, and human resources operations. I observed and contributed to SAP modules such as FICO (Finance and Controlling), Material Management, and Human Capital Management, gaining insights into how enterprise-level applications streamline port activities.

In addition, I actively participated in a software project for circular management, developed using the technologies such as React, Express with MySQL integration. The project aimed to digitalize the internal circular distribution system by creating a role-based portal for uploading, storing, and accessing official documents in PDF format. This included hands-on work with Multer for file upload, axios-based frontend-backend communication, and role-level data filtering.

EXECUTIVE SUMMARY

Jawaharlal Nehru Port, popularly known as Nhava Sheva, is the largest container port in India. Located east of Mumbai in Maharashtra, it is situated on the Arabian Sea and accessed through the Thane Creek. JNPA shares the Mumbai Harbour Channel with Mumbai Port up to the Jawaharlal Nehru Deep Terminal. With a rich operational history, the port was established as an independent facility and has grown to become the nation's leading container port.

The Jawaharlal Nehru Port Authority (JNPA), located in Navi Mumbai, plays a crucial role in India's maritime trade, handling approximately 50% of the country's total containerized cargo at major ports. Operational since 26th May 1989, JNPA has evolved from a bulk cargo terminal into the premier container port in India. It currently ranks 26th among the top 100 container ports globally and maintains connectivity with over 200 ports worldwide.

JNPA operates five key container terminals: Nhava Sheva Free Port Terminals (NSFT), Nhava Sheva International Container Terminal (NSICT), Gateway Terminals India Pvt. Ltd. (GTIPL), Nhava Sheva International Gateway Terminal (NSIGT), and the newly launched Bharat Mumbai Container Terminals Pvt. Ltd. (BMCT). Additionally, the port includes a Coastal Berth to handle coastal cargo, a Shallow Water Berth for general cargo, and a Liquid Cargo Terminal managed by the BPCL-IOCL consortium. A fourth container terminal is currently under development in two phases and is expected to be fully operational by April 2025, adding 2,000 meters of quay length and an annual capacity of 4.8 million TEUs.

For security purposes, the entrance to the server room at PMC is equipped with a card reader and biometric scanner. Fingerprint registration is required for access authorization via the biometric system. Despite the growth of digital communication tools such as internet-based services and email, telephonic communication continues to play a vital role at JNPA. The port carries out extensive work related to telephone exchanges, using a combination of hardware and software systems. The primary technologies include Oracle-based software and SAP for data entry, as well as systems for container transfer management and helpdesk workflows.

Furthermore, the MS Section of JNPA has developed an internal intranet platform designed to share important information across departments. This system serves as a valuable resource for management and facilitates informed decision-making across the organization.

INTRODUCTION (JNPA)

The Jawaharlal Nehru Port Authority (JNPA) in Navi Mumbai is India's Best-performing port, accounting for around 50% of the total containerized cargo volume across the major ports of the country. Commissioned on 26 May 1989, JNPA transformed from a bulk cargo terminal to the leading container port in India in less than three decades. It is the country's first 100% Landlord Major Port. Ranked 26th among the top 100 container ports in the world, JNPA is connected to over 200 ports globally.

Currently, five container terminals operate at JNPA: Nhava Sheva Free Port Terminals (NSFT), Nhava Sheva International Container Terminal (NSICT), Gateway Terminals India Pvt. Ltd. (GTIPL), Nhava Sheva International Gateway Terminal (NSIGT), and the newly commissioned Bharat Mumbai Container Terminals Private Limited (BMCT). The port also features a coastal berth to facilitate coastal cargo, a shallow water berth for general cargo operated by Nhava Sheva Distribution Terminal Pvt. Ltd. (NSDT), and a liquid cargo terminal managed by the BPCL-IOCL consortium. Another liquid cargo terminal developed by JNPA will be operated by M/s JSW - JNPT Liquid Terminal Pvt. Ltd. The second phase of the Fourth Container Terminal is under development and is expected to be fully operational by April 2025, adding an additional quay length of 1,000 meters and an annual capacity of 2.4 million TEUs.

To meet the increasing demand from EXIM trade, JNPA is continuously upgrading its capabilities to offer cost-effective, integrated logistics solutions through infrastructure expansion plans and the implementation of several 'Ease of Doing Business' initiatives. These steps include gate automation, an inter-terminal transfer system linking all terminals, Direct Port Delivery (DPD) services for faster delivery of import consignments at reduced costs, the creation of a logistics data bank to track container movements, and the introduction of new tariff incentives for rail cargo to promote modal shift.

26TH PORT IN THE WORLD

*Source: Lloyd's List One Hundred Containers Ports

In addition, JNPA has also developed a multi-product SEZ in its owned freehold land of 277 hectares to attract international capital and global giants in manufacturing.

In addition, the JNPA is also developing a satellite port at Vadhvan and Dry Ports at Jalna and Wardha, to promote the industrialization of the hinterland.

These expansion plans are being developed in an eco-friendly manner and adequate measures are being taken to maintain ecological balance.

Today nearly 34% of the port area is under green cover including mangroves and the Port has taken up various other initiatives as a Green Port. JNPA has also undertaken various social initiatives to develop a healthy relationship with the port and the local population.

With a view to improving the employable abilities of the local PAPs and other local population, JNPA is in the process of developing – JNPA CIDCO Multi-Skill Development Center to provide and impart proper technical training in different disciplines to PAPs and local persons in order to create suitable skilled labour force by improving/imparting necessary skill to cater to diverse requirements of labour for maritime logistics in Port terminals, requirements of Maritime and other industries, Port Users, Airports and Service Sector.

As India's leading container port, JNPA is constantly upscaling its operations to provide stateof-the-art technology, efficiency and manpower which comply with international standards.

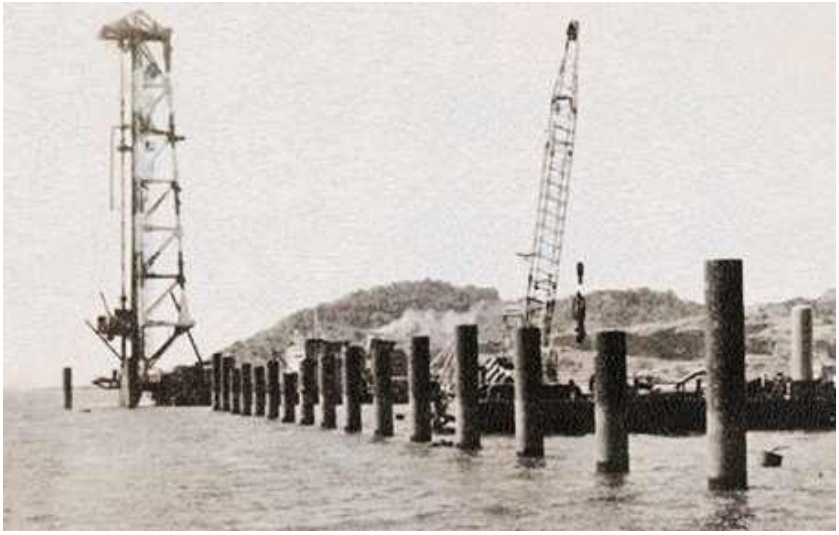
JNPA has firmly anchored itself as the major catalyst for trade and commerce in the country and is strongly committed to providing seamless services to the world that docks here.

History

The port was established on 26 May 1989 and became the second largest and premier container handling port in India,[8][9] surpassing the capacity of 4 million TEUs (Twentyfoot equivalent units) in containers between 2016 and 2021.

JNP consists of a full-fledged customs house, 30 container freight stations and connectivity to 52 inland container depots across the country. The hinterland connectivity both by rail and road is being further strengthened by ongoing projects like the Dedicated Freight Corridor (DFC), which will increase the existing train capacity from 27 to 100 trains per day; MultiModal Logistics Park (MMLP) and widening of the port's connecting roads. Its proximity to the cities of Mumbai, Navi Mumbai and Pune; airports, hotels, exhibition centers, etc. gives the port an edge in efficiency and promptness to address the shippers' needs.

In 2006, the port implemented logistics data tagging of containers.



JNPT houses India's best logistics connectivity with facilitation from container freight stations and logistics parks owned by government as well as private players namely Maharashtra State Warehousing Corporation, DRT, Gateway District Park, Allcargo, Avashya, Continental Warehousing, Ameya Logistics, TransIndia and Contegrate Entrepot. JNPT authorities have been trying to implement direct port delivery model of the containers, a first of its kind model promoted to ease doing business in India.

Nhava Sheva Freeport container terminal

The JNP Trust (JNPT) Container Terminal was originally operated by JNPT. It has a quay length of 680 meters (2,230 ft) with 3 berths.

In 2023 Nhava Sheva Freeport Terminal Private Limited commenced operations of the terminal (renamed to the NSFT terminal) and plans to enhance it to handle 1.8 million TEUs by 2025.

Bharat Mumbai container terminal

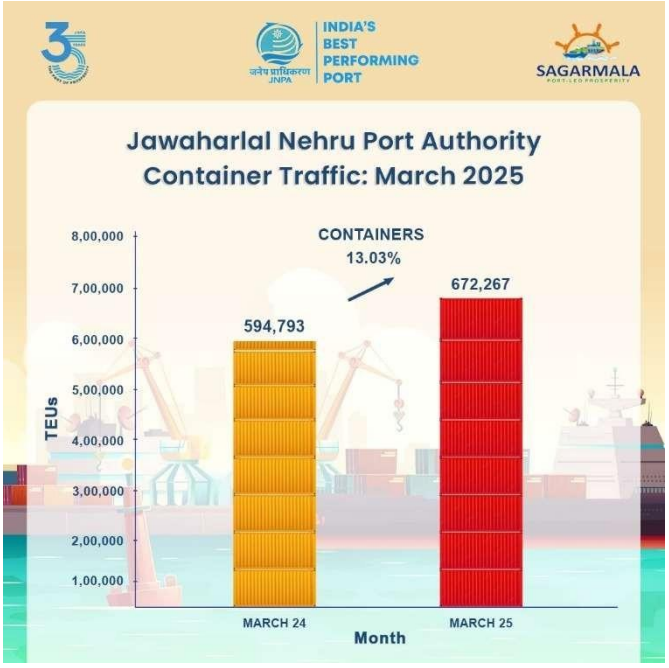
The fourth container terminal, Bharat Mumbai Container Terminal (BMCT), is developed and operated by PSA International. Phase 1 with a capacity of 2.4 million TEUs was completed in Dec 2017. The terminal was planned to have full capacity of 4.8 million TEUs p.a. and a quay length of 2,000 meters with the completion of Phase 2.

Nhava Sheva (India) Gateway container terminal

A new standalone container terminal by the name of NSIGT, having a quay length of 330 meters and a capacity of 0.8 million TEUs was operational in July 2016.

Highlights of March-2025:

- JNPA handled 672,267 TEUs of containers & 8.27 million Tonnes of total cargo in Mar-2025 which is higher by 13.03% & 6.42% respectively as compared to traffic in Mar-2024.
- JNPA has achieved the highest ever monthly Container throughput of 672,267 TEUs in Mar'2025 by surpassing the previous highest of 639,333 TEUs handled in Aug'2024.
- JNPA handled 629 container rakes and 99,929 TEUs during Mar-2025, as compared to 553 rakes and 86,630 TEUs in Mar-2024.
- BMCT has recorded the highest monthly throughput of 2,22,201 TEUs ever handled by any terminal at JNPA.



JNPA has established a multi-product port based Special Economic Zone (SEZ), in line with the objective of port-led industrialization, of Sagarmala vision, under the Ministry of Port, Shipping and Waterways'. This SEZ is strategically located and boasts excellent connectivity via waterways, airways and roadways, making it an ideal hub for export-oriented industries in India. The port has made significant investments in state-of-art infrastructure to support the industries.

JNPA is the developer and the Special Planning Authority for the SEZ, having its own Development Control and Promotion Regulations (DCPR), sanctioned by Government of Maharashtra.

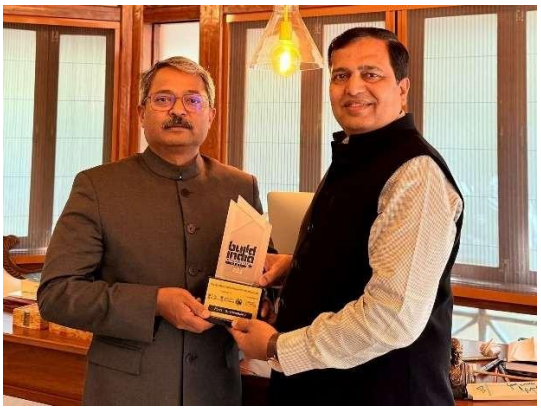
JNPA SEZ
ज ने प प्रा वि आ क्षेत्र
Anchor the Global Trade. Together.

The SEZ covers a total land area of 277.38 hectares (685 acres), with approximately 163 hectares (403 acres) available as leasable land. The SEZ is operational since June 24, 2020. The SEZ allots plots for a 60-year lease period through E-Tender cum E-auction.

As of now, 54 plots have been issued LOI comprising 307 Acres of land, with 9 units and 1 Free Trade Warehousing Zone (FTWZ) already in operation. Other Units/ Co-developers are in different stages of development like awaiting approvals or operational documentation process. The operational units are engaged in diverse sectors such as warehousing, food processing, manufacturing, and trading.

Awards

JNPA is honoured to receive the Certificate of Recognition for being one of the Outstanding Indian Ports at the Sagar Samman Awards on National Maritime Day 2025.



JNPA was honoured with the Build India Infra Award 2025 in the category of PortsSustainability.

JNPA was awarded the Sagar Aankalan Certificate for 1st Rank in Overall Performance in handling Container Cargo < 0.5 million TEUs.

Vadhavan Port Development Project

Introduction

India's maritime sector is undergoing massive expansion under the Maritime India Vision 2030. To strengthen India's competitiveness in global trade and ease the increasing congestion at Jawaharlal Nehru Port Authority (JNPA), the Government of India has approved the development of Vadhavan Port, a world-class greenfield deep-sea port.

The port will be constructed at Vadhavan, Dahanu Taluka, Palghar district, Maharashtra. With a deep natural draft of around 20 meters, it will be capable of accommodating the world's largest container vessels.

The project will be implemented by a Special Purpose Vehicle (SPV) formed by JNPA (majority stakeholder) and Maharashtra Maritime Board (MMB).

Strategic Importance

Complement to JNPA: Vadhavan will reduce cargo burden on JNPA, which is nearing full capacity.

Handling Mega Ships: Accommodates Ultra-Large Container Ships (ULCVs) of 24,000+ TEU.

Boost to Indian Economy: Increases India's share in global container trade.

Logistics Hub: Will transform the Palghar region into a logistics and warehousing hub.

Global Connectivity: Direct connectivity to Europe, Africa, and the Americas via Arabian Sea routes.

Current Status (as of 2025)

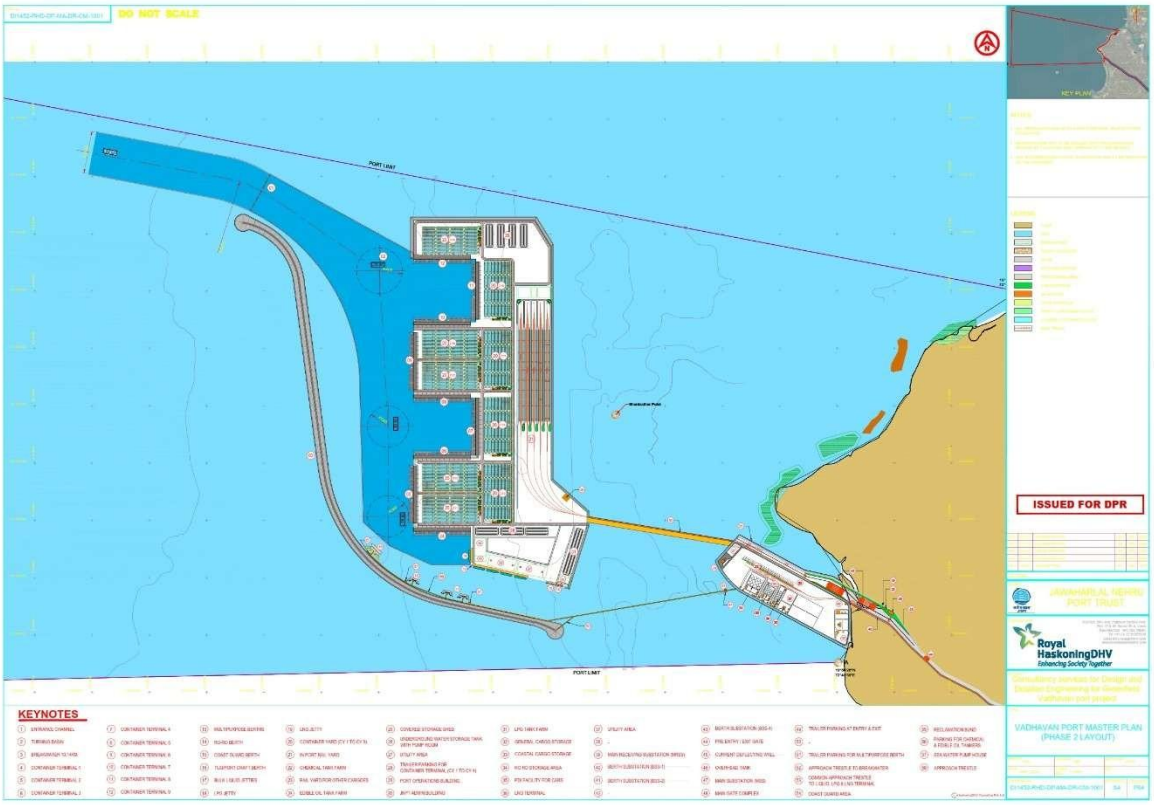
- In-Principle Approval: Received from Cabinet Committee on Economic Affairs (CCEA).
- Detailed Project Report (DPR): Prepared and under review.
- Environment Clearance: Environmental Impact Assessment (EIA) in progress.
- Land Acquisition: Preliminary land surveys and acquisition initiated.
- Expected Construction Start: 2026
- Expected Completion (Phase 1): 2030

Environmental Sustainability Initiatives:

Vadhavan Port is committed to being a Green Port, focusing on:

- Mangrove Conservation: Protective measures for existing ecosystems.
- Shoreline Protection: Modern coastal engineering to prevent erosion.
- Zero Liquid Discharge: No industrial effluent release into sea.
- Renewable Energy Usage: Solar, Wind Energy Integration.
- Green Certification: Aiming for IGBC (Indian Green Building Council) ratings.
- Marine Biodiversity Monitoring: Collaboration with scientific agencies.

Master Plan Layout



Economic and Social Impact:

- Employment Creation:
  - 1.2 lakh (120,000) direct jobs.
  - 2 lakh (200,000) indirect jobs.
- Revenue Generation:
  - Estimated ₹2,000 crore annual revenue from port operations.
- Boost to Local Economy:
  - Small businesses, transportation, hospitality sectors to benefit.
- Skill Development:
  - New training centres for port logistics and management.

Port Layout (Planned)

- Container Terminals (Major)
- Bulk Cargo Terminals
- Liquid Cargo Terminals
- Coastal Shipping Terminals
- Rail Yard with DFC connectivity
- Logistics and Distribution Parks
- Port Service Zones (Administrative buildings, Customs, Immigration)

Skilling and Training Programs

As part of its commitment to inclusive growth and regional development, the Vadhavan Port Project has introduced a comprehensive Skilling Program in collaboration with government and private institutions. The objective is to empower the local population and create a skilled workforce ready for port-related, logistics, and allied industry employment.

- Key Features:

Training Centers: Establishment of Port and Maritime Skill Development Institutes (PMSDIs) near the port site.

- Focus Areas:
  - Port Operations and Management o Crane and Cargo Handling Equipment Training o Logistics and Supply Chain Management o IT Systems in Port Management (SAP, Navis N4) o Electrical & Mechanical Maintenance o Maritime Safety and Environmental Compliance

- Partnerships:
  - In collaboration with Skill India Mission, NSDC (National Skill Development Corporation), and JNPA's own HRD initiatives.
  - Engagement with nearby Industrial Training Institutes (ITIs) and Polytechnic Colleges in Maharashtra.

- Target Groups:
  - Youth and unemployed individuals from Palghar, Dahanu, and nearby rural areas
  - Women empowerment initiatives through basic computer and logistics training

- Certification: Government-recognized certificates enabling job placement within Vadhavan Port or other Indian ports.

- Outcome Goal: To train 25,000+ people over 5 years before and during the port commissioning phase.

JNPA was delighted to host principals and vice principals from colleges in Palghar. Our Chairperson Shri Unmesh Sharad Wagh, IRS, CMD, VPPL warmly welcomed them.

JNPA's Vadhvan Port Project Limited (VPPL) is looking forward to collaborate with these colleges to upskill students from the Palghar district through various courses registered under the Vadhvan Skilling Program.

The visiting delegation also toured the port, gaining a deeper understanding of the maritime sector and its importance for economic growth.

Unlock Your Potential with Vadhvan Skilling Programs!

Join 18 skill-enhancing courses fully funded by VPPL under Vadhvan Skilling program designed to help you upskill and grow. This initiative aims to provide a chance to the local youth to gain valuable skills and employment opportunities at completely free cost!

Simply scan the QR code and register yourself in the Vadhvan Skilling Programs.

ADMINISTRATIVE BUILDING

Office administration is a set of day-to-day activities that are related to financial planning, record keeping &billing, personnel, physical distribution and logistics, within an organization.

Office administrator has the responsibility of ensuring that the administrative activities within an organization run efficiently, by providing structure to other employees throughout the organization.

These activities range from being responsible for the management of human resources, budgets and records, to undertaking the role of supervising other employees. This is a function of an Administrative building at JNPA.

These responsibilities can vary depending on the employer and level of education.

Role of Management Services Section

The Management Services (MS) section plays a crucial role in delivering Information Technology (IT) solutions that support the efficient and smooth functioning of port operations.

As a part of the Finance Department, the MS section is responsible for overseeing all computer-controlled operations, communication systems, and various computerized management functions essential for the day-to-day activities of the port.

The section is tasked with the operation and maintenance of critical IT infrastructure, including Electronic Data Interchange (EDI), video conferencing systems, the port's official website (both Intranet and Internet), mail messaging services, and Inmarsat-based communication systems. It also handles data and voice communication networks to ensure uninterrupted connectivity across departments.

To effectively manage these services, the MS section operates two major computer centers — one located within the port premises and the other at the Administration Building. These centers are networked to enable seamless and secure communication and data exchange throughout the organization.

In addition to these, the MS section also takes care of the maintenance and support of personal computers (PCs) used across the port.

Sites at Management Services

E-Tender:

GeMs: It is the dedicated E market for different goods and services procured by Government/ Organizations/ Departments/ PSUs.

Government e Marketplace (GeM), created in a record time of five months, facilitates online procurement of common use Goods & Services required by various Government/ Organizations/ Departments/ PSUs.

GeM aims to enhance transparency, efficiency and speed in public procurement. It provides the tools of e-building, reverse auction and demand aggregation to facilitates the government users, achieve the best value for their money.

E-Procure: It is the E-tender site are handled at the management services department by the employees and the site is maintained by the software engineers.

The E-Procure site is advancement to the offline mode of tender passing which is sometimes defined to be biased. The online procedure removes the biasness and provides unbiased selection of the best technically and economically fit tenders.

The site provides latest updates on JNPA Tenders, eProcurement Tenders and other online tender notices published by Jawaharlal Nehru Port Authority (JNPA).

It is the online tender website which provides unlimited access to JNPA Tenders and other procurement notices types like: open tender, public tender, online tenders limited from Jawaharlal Nehru Port Authority (JNPA).

Users can Register and search tenders, view tender details and download tender document from Jawaharlal Nehru Port Authority (JNPA)

The e-tender involves the following process:

Creation: Tenders are created by keeping in mind the needs and requirements w.r.t economy.

Publication: The tenders generated are uploaded to the site. Evaluation:

Includes two stages:

Technical: Only those tenders that technically matching with the requirements are selected.

Financial: The tenders that covers the requirement and provides the best rate are selection.

AOC (Award of Contract): The final selected tender requires the approval of higher authority. E-Office:

E-office is similar to the mailing system used daily. The e-office site is made for managing the files in office using online facilities.

It provides platform to save the soft copy of files and transfer the files from one person to other over the internet.

It is safe and secure for the sharing of files. It removes the process of unnecessary piling of paper, and improves the quality of work being time efficient.

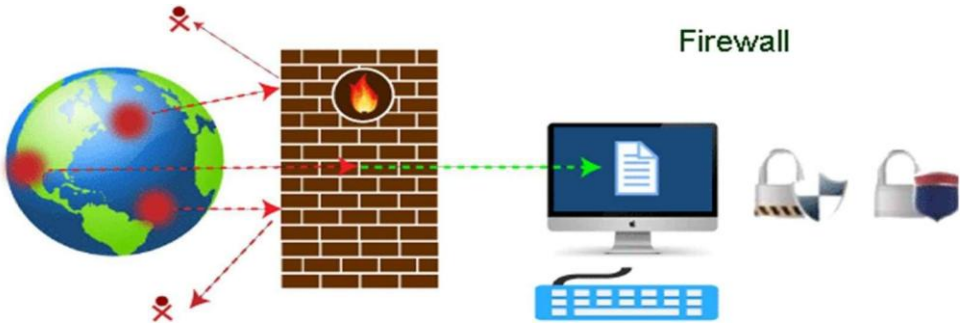
The files that are sent over the internet can be signed digitally by the respective authority. Hence there is no waste of time and paper. It is efficient at times of urgency.

Firewall and types of firewall

Firewall:

A firewall can be defined as a special type of network security device or a software program that monitors and filters incoming and outgoing network traffic based on a defined set of security rules. It acts as a barrier between internal private networks and external sources (such as the public Internet).

The primary purpose of a firewall is to allow non-threatening traffic and prevent malicious or unwanted data traffic for protecting the computer from viruses and attacks. A firewall is a cybersecurity tool that filters network traffic and helps users block malicious software from accessing the Internet in infected computers.



How does a firewall work?

A firewall system analyzes network traffic based on pre-defined rules. It then filters the traffic and prevents any such traffic coming from unreliable or suspicious sources. It only allows incoming traffic that is configured to accept. Typically, firewalls intercept network traffic at a computer's entry point, known as a port. Firewalls perform this task by allowing or blocking specific data packets (units of communication transferred over a digital network) based on predefined security rules. Incoming traffic is allowed only through trusted IP addresses, or sources.

Types of Firewalls

Depending on their structure and functionality, there are different types of firewalls. The following is a list of some common types of firewalls:

- Proxy Firewall
- Packet-filtering firewalls
- Stateful Multi-layer Inspection (SMLI) Firewall
- Unified threat management (UTM) firewall

SonicWall Firewall

SonicWall firewalls are next-generation network security appliances that offer comprehensive protection against a wide range of cyber threats. They are known for their advanced threat prevention capabilities, including deep packet inspection (DPI), intrusion prevention system (IPS), and malware detection. SonicWall firewalls also provide features like application control, content filtering, and VPN for secure remote access.

Key Features and Capabilities:

- Advanced Threat Prevention:

SonicWall firewalls utilize deep packet inspection (DPI) to analyze all network traffic, including encrypted traffic, for malicious activity.

- Intrusion Prevention System (IPS):

They include an IPS that blocks known malicious attacks, providing a layer of defense against various cyber threats.

- Malware Protection:

SonicWall firewalls detect and block malware, including ransomware, viruses, and other malicious software.

- Application Control:

They provide visibility and control over network applications, allowing administrators to manage and prioritize traffic based on application type.

- Content Filtering:

SonicWall firewalls can filter web content based on keywords, categories, or websites, helping to prevent users from accessing malicious or inappropriate content.

- VPN:

They offer secure VPN tunnels for remote users, enabling them to access the network securely from any location.

- Real-time Threat Handling:

SonicWall firewalls can detect and respond to real-time threats, providing immediate protection against evolving attacks.

- Centralized Management:

SonicWall provides centralized management tools for managing multiple firewalls, simplifying administration and improving security.

- Analytics and Reporting:

They offer detailed analytics and reporting features, providing insights into network traffic, threats, and security performance.

SonicWall Firewall Series:

- TZ Series:

Designed for small to medium-sized businesses and branch offices, offering a balance of security and performance.

- NSA Series:

A more robust series for mid-sized and larger businesses, providing advanced threat prevention and high-speed performance.

- NSsp Series:

A high-end series for large enterprises, data centres, and service providers, offering the highest levels of security and performance.

- NSv Series:

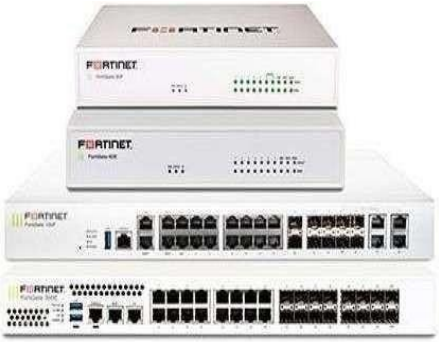
A virtual firewall series that offers the same security capabilities as physical firewalls but with the flexibility and scalability of virtualization.

FortiNet

FortiGate NGFWs deliver industry-leading enterprise security for any edge at any scale with full visibility and threat protection.

Organizations can weave security deep into the hybrid IT architecture and build security driven networks to achieve:

- Ultra-fast security, end to end
- Consistent real-time defense with FortiGuard Services
- Excellent user experience with security processing units
- Operational efficiency and automated workflows FortiGate NGFWs enable organizations to build high-performance, ultra scalable, and security-driven networks



My Role as an Intern

During the period from 9th June 2025 to 8th July 2025, I had the privilege of competing in an internship at JNPA, Sheva.

This report provides an overview of the tasks accomplished, Skills acquired, and contribution made for 1 month Internship duration.

- Maintenance of Computer Systems
- Windows Installation
- Joining the Domain
- Changing the IP Address
- Application Installation
- Data Recovery
- Sessions and learnings
- Created a Database in MySQL Server
- Created a Circular Management system.

Maintenance of Computer Systems

Maintenance of computers was going on in whole JNPA port which created great opportunity for me to learn various things. Firstly, the sir told me all the information about the computer and its operation.

All the computers from the port were taken to the M.S section one by one to replace them with the new ones. As the old computers contained various important data , the data was transferred to the new PC and were sent to the respected offices in the port.

Windows Installation

As part of the computer replacement process at JNPA, it was necessary to install an operating system on all the new PCs. The port primarily uses Windows as its operating system, so Windows was installed on all systems. Some of the PCs were still running Windows 8, which had reached its end of support from Microsoft. To keep the systems up to date and secure, we upgraded those PCs to Windows 10.

This ensured that all systems were running the latest, supported version of the operating system.

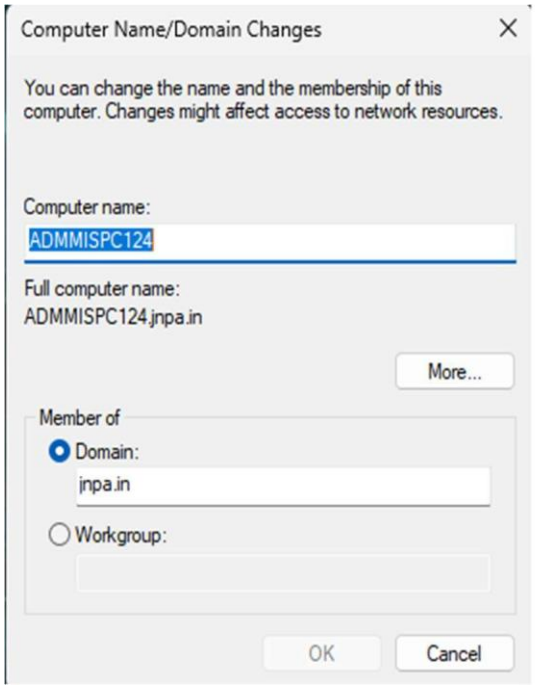
Joining the Domain

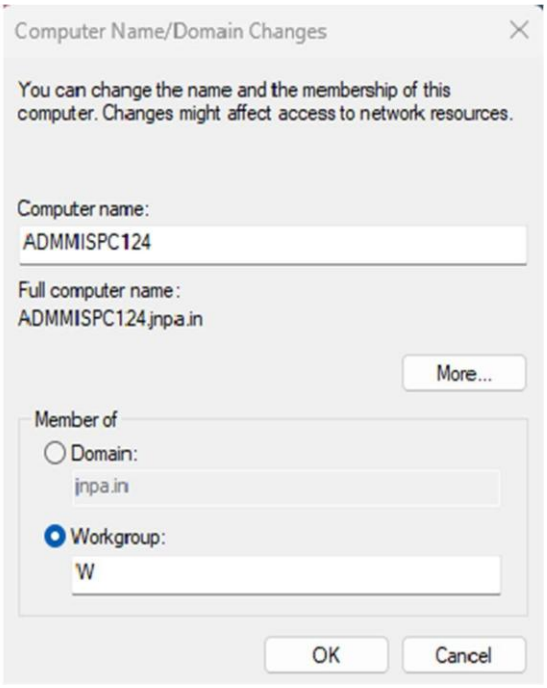
A domain, in the context of networking, refers to any group of users, workstations, devices, printers, computers and database servers that share different types of data via network resources.

Here we learned how to join a PC to a domain.

To join a computer to a domain

1. On the Start screen, type Control Panel, and then press ENTER.
2. Navigate to System and Security, and then click System.
3. Under Computer name, domain, and workgroup settings, click Change settings.
4. On the Computer Name tab, click Change.
5. Under Member of, click Domain, type the name of the domain that you wish this computer to join, and then click OK.
6. Click OK and then restart the computer.

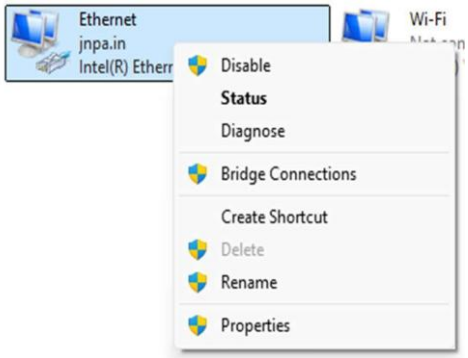


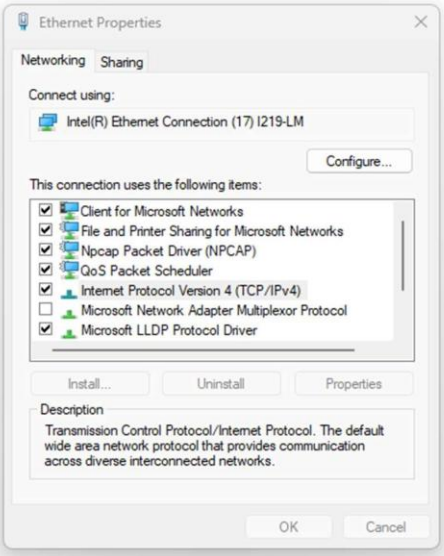


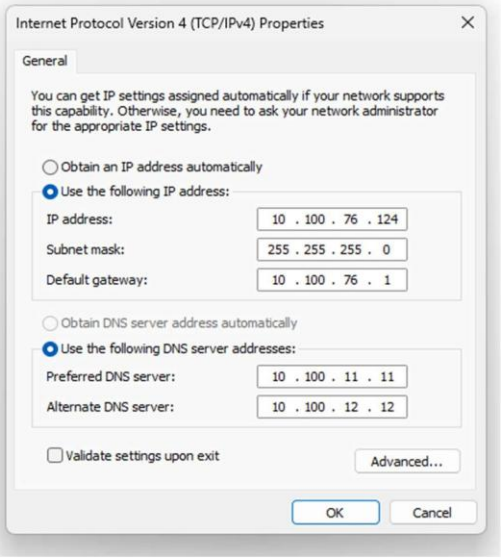
Changing the IP Address

A unique string of characters that identifies each computer using the Internet Protocol to communicate over a network.

1. Open Control Panel
2. Network and Sharing Centre
3. Change adapter settings
4. Select Properties
5. Select Internet Protocol Version 4 (TCP/IPv4).
6. Either alter the IP address or select Obtain an IP address automatically to have the router control the IP address.
7. If you leave the setting to obtain an IP address automatically, the router automatically assigns an IP address to the computer. However, when you enter the IP address information, the computer's IP address stays static.


Ethernet
jnpa.in
Intel(R) Ether
Wi-Fi
Disable
Status
Diagnose
Bridge Connections
Create Shortcut
Delete
Rename
Properties


Ethernet Properties
Networking
Sharing
Connect using:
Intel(R) Ethernet Connection (17) I219-LM
Configure...
This connection uses the following items:
Client for Microsoft Networks
File and Printer Sharing for Microsoft Networks
Npcap Packet Driver (NPCAP)
QoS Packet Scheduler
Internet Protocol Version 4 (TCP/IPv4)
Microsoft Network Adapter Multiplexor Protocol
Microsoft LLDP Protocol Driver
Install...
Uninstall
Properties
Description
Transmission Control Protocol/Internet Protocol. The default wide area network protocol that provides communication across diverse interconnected networks.
OK
Cancel


Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 100 . 76 . 124
Subnet mask: 255 . 255 . 255 . 0
Default gateway: 10 . 100 . 76 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server: 10 . 100 . 11 . 11
Alternate DNS server: 10 . 100 . 12 . 12
Validate settings upon exit
Advanced...
OK
Cancel

Software Installation

In addition to installing the operating system, various specialized applications were required for use at JNPA.

These applications, such as SAP, Navis N4, and e-Office, are essential tools for the daily operations of JNPA employees and are specifically tailored to their work needs.

After completing the fresh installation of Windows 10, we moved on to install necessary software for general use, including Google Chrome for web browsing, Adobe Acrobat Reader for viewing PDFs, and Microsoft Office for word processing, spreadsheets, and presentations.

These software installations ensured that the systems were fully functional and ready for use in the work environment at JNPA

Data Recovery

The old PCs at JNPA contained important user data stored on their hard drives, which needed to be transferred to the new systems.

The old PCs used traditional CPU units, while the new ones were all-in-one PCs, which are lighter, more compact, and easier to carry.

Before proceeding with the installation of Windows 10 on the new PCs, we first manually transferred the user data from the old PCs to an external SSD. This was done using a converter to ensure a smooth data transfer process. Once the data was securely copied, we continued with the installation of the operating system and other necessary steps to set up

Trellix Agent

For updating or installing Trellix, begin by pressing Windows + R, then type \\pmccshare1 and press Enter. Navigate to the software folder, then open the trellix agent folder. Inside, locate the file named framepkg, right-click on it, and select Run as administrator. Enter the administrator password when prompted.

Before proceeding, ensure the PC is part of a domain. If it is not, join the PC to the domain. Once the PC is in the domain, press Windows + R again, type \\pmccshare1, and extract the necessary files. If a password prompt appears, use the credentials provided by the support team.

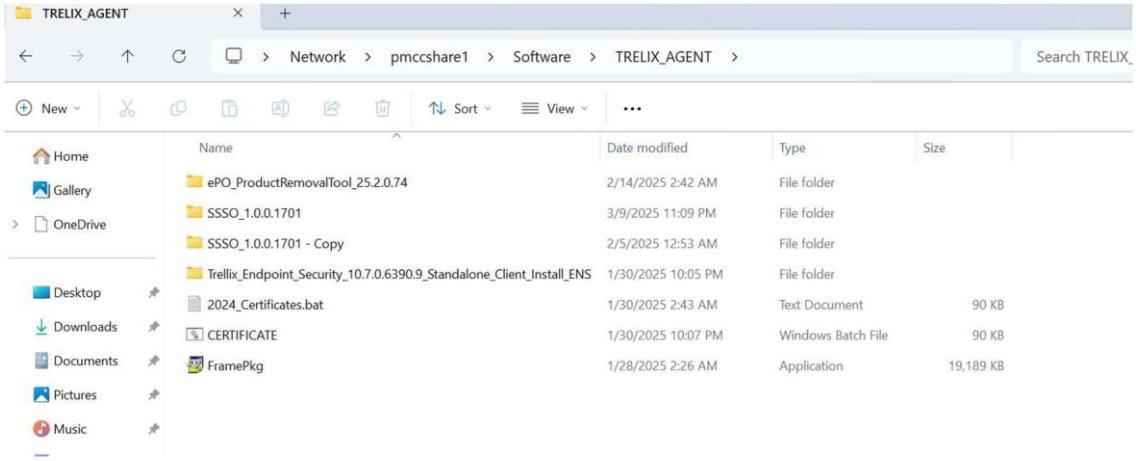
After completing the above steps, go to the Trellix icon in the taskbar and update the security settings. Make sure that the Trellix software is updated to the latest version, which should be 5.8.3.602.

If Trellix is still not functioning correctly after the update, proceed with the firewall installation. Go to \pmccshare1 → software → trellix agent, then open the trellix endpoint security folder and locate the setup file. Right-click on it and select Run as administrator. Follow the prompts to install the software. Once the installation is complete, go to the Trellix icon in the taskbar and click on About to confirm that the installation and update have been successfully completed.

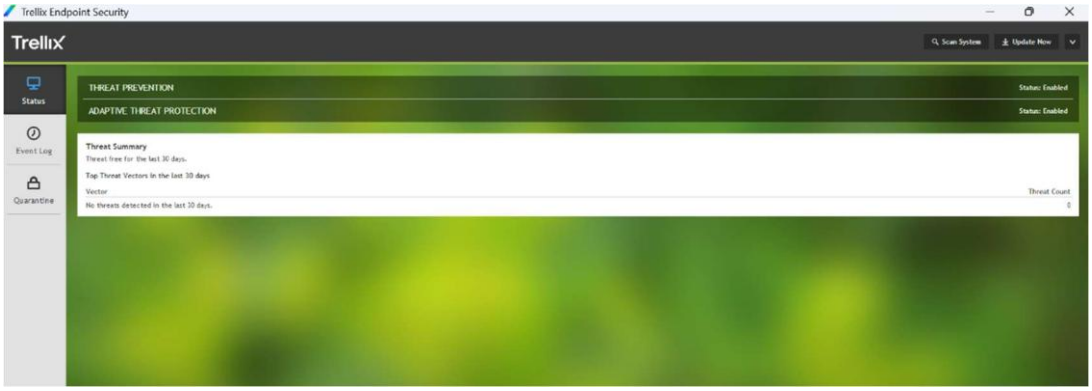
Trellix

ePO

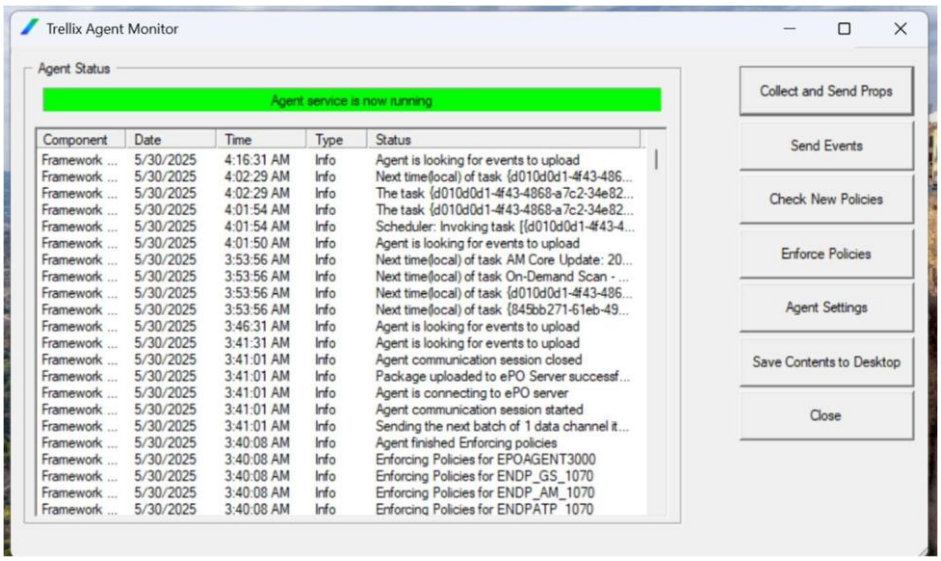
Network > pmccshare1 > Software > Trellex_agent



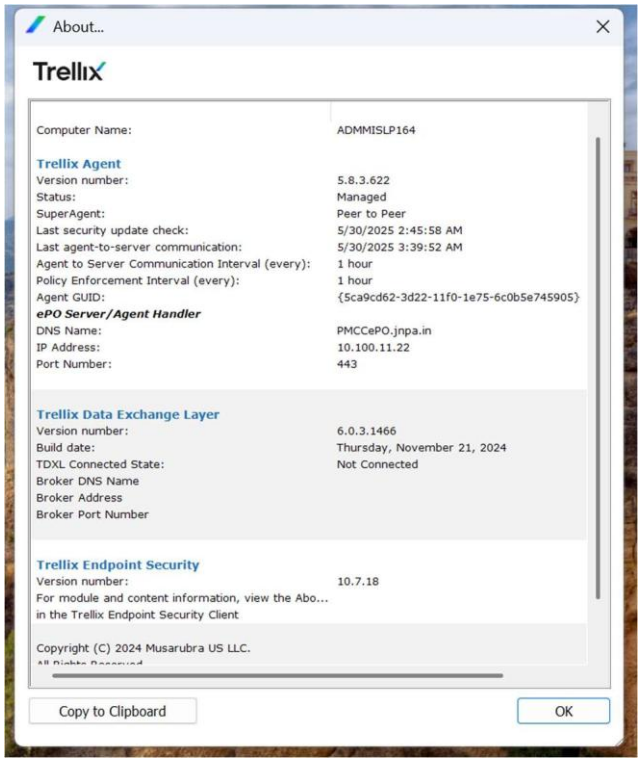
Trellix endpoint security >

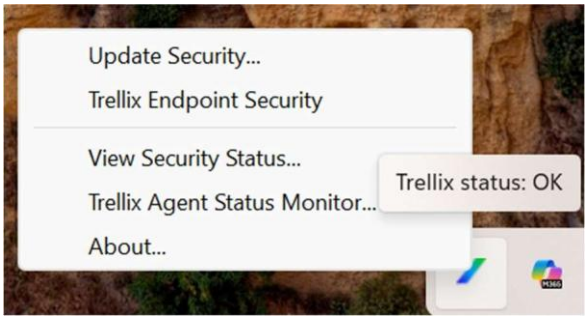


Trellix Agent Monitor:



Component	Date	Time	Type	Status
Framework ...	5/30/2025	4:16:31 AM	Info	Agent is looking for events to upload
Framework ...	5/30/2025	4:02:29 AM	Info	Next time(local) of task {d010d0d1-4f43-486...
Framework ...	5/30/2025	4:02:29 AM	Info	The task {d010d0d1-4f43-4868-a7c2-34e82...
Framework ...	5/30/2025	4:01:54 AM	Info	The task {d010d0d1-4f43-4868-a7c2-34e82...
Framework ...	5/30/2025	4:01:54 AM	Info	Scheduler: Invoking task [{d010d0d1-4f43-4...
Framework ...	5/30/2025	4:01:50 AM	Info	Agent is looking for events to upload
Framework ...	5/30/2025	3:53:56 AM	Info	Next time(local) of task AM Core Update: 20...
Framework ...	5/30/2025	3:53:56 AM	Info	Next time(local) of task On-Demand Scan - ...
Framework ...	5/30/2025	3:53:56 AM	Info	Next time(local) of task {d010d0d1-4f43-486...
Framework ...	5/30/2025	3:53:56 AM	Info	Next time(local) of task {845bb271-61eb-49...
Framework ...	5/30/2025	3:46:31 AM	Info	Agent is looking for events to upload
Framework ...	5/30/2025	3:41:31 AM	Info	Agent is looking for events to upload
Framework ...	5/30/2025	3:41:01 AM	Info	Agent communication session closed
Framework ...	5/30/2025	3:41:01 AM	Info	Package uploaded to ePO Server successf...
Framework ...	5/30/2025	3:41:01 AM	Info	Agent is connecting to ePO server
Framework ...	5/30/2025	3:41:01 AM	Info	Agent communication session started
Framework ...	5/30/2025	3:41:01 AM	Info	Sending the next batch of 1 data channel it...
Framework ...	5/30/2025	3:40:08 AM	Info	Agent finished Enforcing policies
Framework ...	5/30/2025	3:40:08 AM	Info	Enforcing Policies for EPOAGENT3000
Framework ...	5/30/2025	3:40:08 AM	Info	Enforcing Policies for ENDP_GS_1070
Framework ...	5/30/2025	3:40:08 AM	Info	Enforcing Policies for ENDP_AM_1070
Framework ...	5/30/2025	3:40:08 AM	Info	Enforcing Policies for ENDPATP_1070





Sessions and learnings

1. Networking Concepts & Infrastructure

During the internship, we were introduced to the network infrastructure used in a professional enterprise environment. We studied the architecture through a real-world example, including a detailed network diagram involving multiple branches, firewalls, MPLS, and internet links.

One key concept covered was Active Directory (AD), which serves as a centralized system to manage all computers and users within a network. It allows administrators to set user permissions, define access policies, and monitor connected desktops from a single dashboard. This functions like a control unit that governs how users interact with the system and enforces security protocols.

We also learned about DNS (Domain Name System), which translates domain names like google.com into IP addresses. It was explained how the local DNS server first receives the request, resolves the IP, and then routes the traffic. In the organizational setup, custom configurations were implemented so internal domains were resolved statically to manage services within the same domain name.

Another important concept introduced was Hyper-V, a virtualization platform used to create and manage multiple virtual machines (VMs). Each VM was configured with its own dedicated RAM and storage. We were informed that system specifications of VMs (like RAM or disk space) can only be changed while the machine is turned off, maintaining system stability.

Firewall systems were also thoroughly explained. The organization used a dualfirewall setup to enhance security. Most network traffic is blocked by default, and only explicitly defined rules allow certain data flows. This helps protect internal systems from external threats.

In the PMCC–Mumbai architecture, we saw that not all branches maintain their own server infrastructure. Some branches, like the Express Tower in Mumbai, rely on centralized servers located at PMCC and connect through secure MPLS links. Furthermore, the IT team leverages centralized update management tools to push operating system updates to all connected machines. These updates can be scheduled and monitored via dashboards, which display the status of each machine. In case of failures or glitches, updates are manually pushed.

Lastly, we were introduced to redundancy planning in the server room. Systems like UPS (Uninterruptible Power Supply), power supply, network switches and storage are all backed by redundant systems to prevent downtime and ensure smooth operations.

2. Cybersecurity Awareness and Practices

As part of the internship, we were given an in-depth overview of cybersecurity concepts and real-world practices used to protect organizational networks and data from internal and external threats. The sessions built upon foundational ethical hacking concepts and extended them into practical corporate scenarios. We learned about common types of cyber-attacks, especially those targeting authentication systems:

- Brute Force Attacks – where automated tools try all possible combinations of passwords to gain access. These attacks are especially effective against weak or reused passwords.
- Dictionary Attacks – similar to brute force but based on a list of commonly used passwords or leaked credentials.
- Phishing & Social Engineering – techniques used to trick users into revealing sensitive information (like passwords or OTPs) by pretending to be a trusted source (emails, fake login pages, calls).
- Man-in-the-Middle (MITM) Attacks – where an attacker intercepts communication between two parties, often used to steal data or credentials.
- Keylogging – where a malware records keystrokes to capture passwords or sensitive data.

To counter these threats, various security precautions and best practices were discussed:

- Enforcing strong password policies: minimum length, use of uppercase/lowercase letters, numbers, and symbols.
- Implementing two-factor authentication (2FA) or multi-factor authentication (MFA) to add an extra layer of protection.
- Limiting login attempts to block repeated failed logins and mitigate brute-force attacks.
- Educating employees through awareness programs to recognize phishing attempts and suspicious activity.
- Using secure protocols (HTTPS, SSH) for communication and file transfers.
- Ensuring all endpoints are protected with antivirus/anti-malware tools and regular system updates.

- Firewalls – configured to block unauthorized traffic and only allow permitted protocols and IPs.
- Intrusion Detection and Prevention Systems (IDPS) – used to monitor network traffic and flag potential threats.
- Access control policies – limiting user access based on role (principle of least privilege).
- Data encryption – both at rest and in transit to ensure confidentiality even if data is intercepted.
- Security audits and vulnerability assessments – to find and patch weak points in the infrastructure.
- Digital Forensics & Incident Response – the process of identifying, investigating, and mitigating security breaches.
- One of the key practices discussed was the "clear screen policy", which mandates that employees must lock their computers when stepping away from their desks.
- This ensures that confidential or sensitive data on-screen is not exposed to unauthorized individuals.

- In case an employee forgets to manually lock their system, it was recommended to configure an auto-lock timeout (typically 3–5 minutes of inactivity), which automatically secures the workstation by requiring login credentials again.

- Other good practices shared during the session included:

- Avoiding password sharing: Employees should never share login credentials with colleagues or write them down in visible places.
- Using screen privacy filters: In open office environments, privacy filters can prevent shoulder surfing and keep sensitive on-screen data private.
- Regular system updates: Employees are advised to keep their systems updated to patch known vulnerabilities and reduce the risk of exploitation.
- These simple but effective practices are crucial in maintaining the cybersecurity posture of any organization, particularly in environments where sensitive information is handled. The session highlighted that cybersecurity is not only about tools and software, but also about consistent human behaviour and awareness.

3. Oracle Financial System Overview

During the Oracle training session, we were introduced to the Oracle Financial Accounting System, a powerful ERP solution widely used by enterprises to manage financial operations. The module is designed to handle large-scale financial transactions and maintain accurate, real-time accounting records for informed business decisions.
We learned that the General Ledger (GL) is at the heart of the financial accounting system. It records and organizes data related to assets, liabilities, income, and expenditure, forming the foundation for preparing critical financial reports.

Key Modules Covered:

- Revenue (Bulk & Tank Forms): Used to manage and record revenue from bulk goods or tank storage — common in logistics and industrial sectors.
- Cash and Bank Management: Monitors all cash inflows/outflows and bank transactions to ensure liquidity and timely reconciliation.
- Expenditure Management: Tracks and controls company spending, ensuring expenses align with budgets.
- Trial Balance: A summary report ensuring total credits equal total debits; includes income, expenditure, assets, and liabilities.
- Profit & Loss Statement (Income vs. Expenditure): Provides insights into company profitability over a set period.
- Works Management: Oversees project-related financial tracking for departmental or field operations.
- Materials Management: Manages procurement, usage, and costing of materials used across departments.
- Estate Rentals: Records rental income from organizational property or land leases.
- Establishment: Handles employee-related financial data like salary structure, benefits, etc.
- Administration: Provides system-level configurations, access control, and role management.
- Asset Management: Tracks the acquisition, depreciation, and retirement of physical assets.
- Budget Management: Assists in planning, allocating, and monitoring budget limits across departments.
- Investment Management: Keeps records of organization's financial investments and returns.
- Debts Management: Manages borrowed funds, loans, and their repayment tracking.

4. Computer Hardware and System Components

As part of the internship, we attended a session on computer hardware, where we explored the physical components of a CPU and how they interact to form a functional system. The session covered both the theoretical and practical aspects of internal hardware, with a focus on how power flows and how components communicate.

We began by understanding the flow of AC (Alternating Current) and its conversion into DC (Direct Current) through a device called the SMPS (Switched Mode Power Supply). The SMPS is responsible for supplying regulated voltage to the internal components of the computer. It uses rectifiers and capacitors to stabilize and filter the voltage. Capacitors store energy and help smooth out power fluctuations, ensuring stable performance.

The session then introduced different motherboard topologies, such as bus and star configurations, which define how components like RAM, CPU, and I/O devices are interconnected. We also discussed ports and interfaces commonly used in modern systems, such as USB, HDMI, and Ethernet, which allow external devices to communicate with the CPU.

A key hands-on discussion was about the motherboard — the central circuit board in the CPU cabinet. We learned that if the motherboard is removed or disconnected, the CPU emits a sound, indicating a hardware malfunction or missing component. Reconnecting the motherboard resolves this, showing how essential it is for communication between all hardware units.
Additionally, we covered:

- RAM and ROM differences — with RAM handling temporary storage and ROM storing permanent boot instructions.
- Storage devices: SSDs and HDDs, with SSDs being faster and more reliable.
- CPU architecture basics including the ALU (Arithmetic Logic Unit) and Control Unit, which execute and manage instructions.

- Importance of cooling systems like fans and heat sinks to prevent overheating. This session helped us understand the core physical structure of a computer, from power supply to communication ports, and provided insight into diagnosing basic hardware issues using visual cues or system beeps.

5. Oracle HRMS & SAP

As part of the internship training, we were introduced to Oracle Human Resource Management System (HRMS) and SAP S/4HANA, two widely used enterprise applications for managing employee data, payroll, and administrative workflows.

These sessions focused on understanding how large organizations handle employee-related information in a structured and secure way.

Oracle HRMS (Human Resource Management System)

The Oracle HRMS session demonstrated how personal, employment, and payroll data of employees is managed within the system.

The HR Master Data screen allowed us to view and update employee records across various categories such as:

- Personal details (name, ID, personnel area/subarea)
- Organizational assignment
- Address and contact information
- Bank details
- Family and dependents
- Payroll planning and attendance tracking

We learned how the system is used to maintain accurate employee records and facilitate HR operations like updating contracts, tracking attendance, and managing payroll data. These features ensure centralized control and help reduce errors in manual data handling.

SAP S/4HANA (ERP – Enterprise Resource Planning)

The SAP session focused on SAP S/4HANA, particularly within the HR and payroll modules. We observed the SAP Easy Access menu, which provides navigation to various features such as:

- Displaying or editing HR Master Data
- Managing material and time records
- Running reports and queries related to HR data
- Accessing the Production System interface used by actual HR/Admin staff

The SAP module promotes secure handling of sensitive data, ensures role-based access, and allows HR personnel to manage employee lifecycles from recruitment to retirement.

Payroll Processing System

The session also introduced us to a dedicated Payroll Processing System, which handled various aspects of salary and benefits management. Key modules included:

- Personnel Information
- Attendance and Leave Maintenance
- Loan and GPF Account Management
- Payroll Calculation
- Bonuses and Salary Budgeting
- Pension Management
- Training Records

This system automates monthly salary processing, deductions, bonuses, pension contributions, and generates salary slips. It ensures accuracy, reduces manual effort, and complies with organizational financial policies.

SQL commands introduced:

1. DML – Data Manipulation Language

Used to manipulate data in existing tables.

Command	Description	Example
SELECT	Retrieves data from a table	SELECT * FROM employees;
INSERT	Adds new records to a table	INSERT INTO employees (id, name) VALUES (1, 'Alice');
UPDATE	Modifies existing records	UPDATE employees SET name = 'Anu' WHERE id = 1;
DELETE	Deletes existing records	DELETE FROM employees WHERE id = 1;

2. DDL – Data Definition Language

Used to define or modify table structure.

Command	Description	Example
CREATE	Creates a new table or database	CREATE TABLE students (id INT, name VARCHAR(50));
ALTER	Modifies an existing table	ALTER TABLE students ADD age INT;
DROP	Deletes a table or database	DROP TABLE students;
TRUNCATE	Removes all rows from a table (faster than DELETE)	TRUNCATE TABLE students;
RENAME	Renames a table	RENAME TABLE students TO alumni;

3. DCL – Data Control Language

Used to control access and permissions.

Command	Description	Example
GRANT	Gives privileges to users	GRANT SELECT, INSERT ON employees TO alice;
REVOKE	Removes privileges from users	REVOKE INSERT ON employees FROM alice;

4. TCL – Transaction Control Language Used to manage transactions in SQL.

Command	Description	Example
COMMIT	Saves all changes made in the transaction	COMMIT;
ROLLBACK	Undoes changes since the last COMMIT	ROLLBACK;
SAVEPOINT	Sets a point to rollback to later	SAVEPOINT save1;
RELEASE SAVEPOINT	Deletes a savepoint	RELEASE SAVEPOINT save1;

Clause usage:

Clause	Usage Example
WHERE	SELECT * FROM employees WHERE age > 25;
ORDER BY	SELECT * FROM students ORDER BY name ASC;
GROUP BY	SELECT dept, COUNT(*) FROM employees GROUP BY dept;
HAVING	SELECT dept, COUNT(*) FROM employees GROUP BY dept HAVING COUNT(*) > 2;
IN / BETWEEN	SELECT * FROM students WHERE id IN (1,2,3);

Circular Management System for Departmental Communication

(Developed during internship)

Overview :

The Circular Management System is a centralized web-based application developed to streamline the process of uploading, managing, and viewing official circulars within an organization or institution. The system is designed to be role-based, offering different functionalities for Admin, Department Heads, and General Viewers.

Objectives:

- Digitize circular distribution
- Centralize management-
- Role-based access
- Easy search and filtering
- Maintain timestamped log

Key Modules :

A. Authentication and Role-Based Access

- Admin: All access
- Department Head: Upload/view own department
- Viewer: Read-only access

B. Circular Upload Interface

- Title, Department, Date, Description, PDF file

C. View Circulars

- Filter by department
- Embedded viewer (no new tab)

D. PDF Management

- Secure upload via multer

Technology Stack:

- Frontend: React.js, HTML, CSS, JS
- Backend: Node.js, Express
- Database: MySQL
- Upload: Multer
- Storage: Local server

System Implementation :

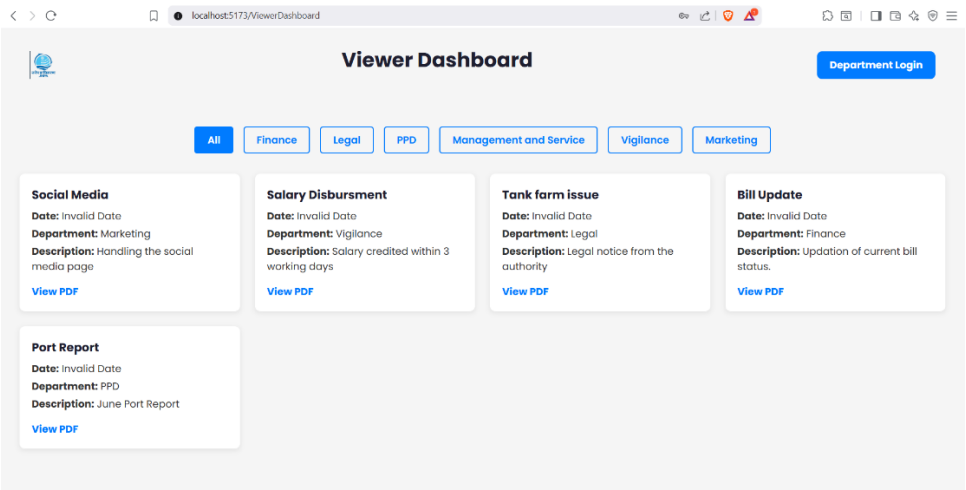


Fig.1-Viewer Dashboard

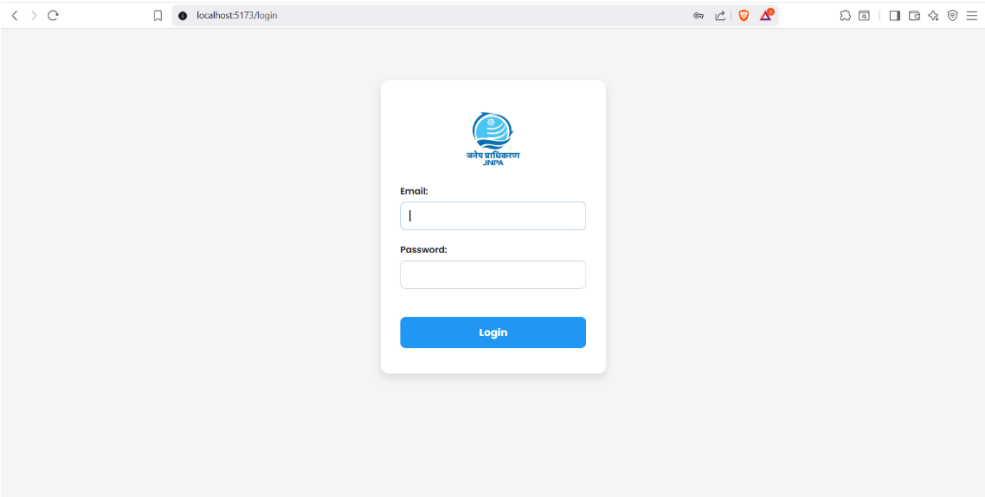


Fig.2 - Login

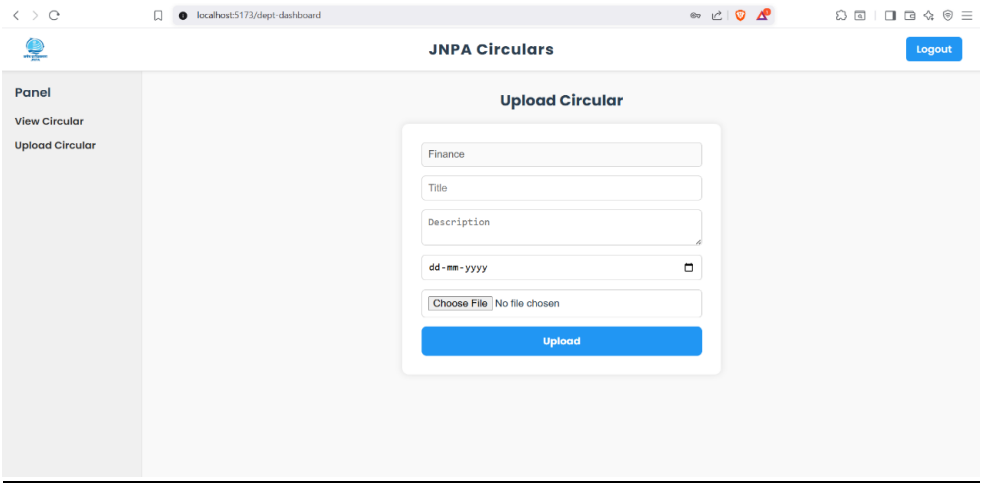


Fig.3 – Department Dashboard

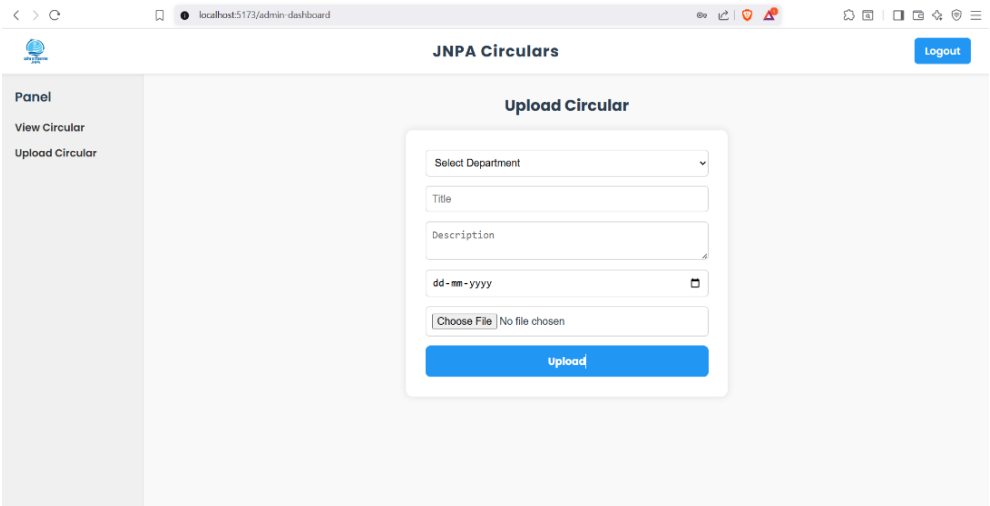


Fig.4 – Admin Dashboard

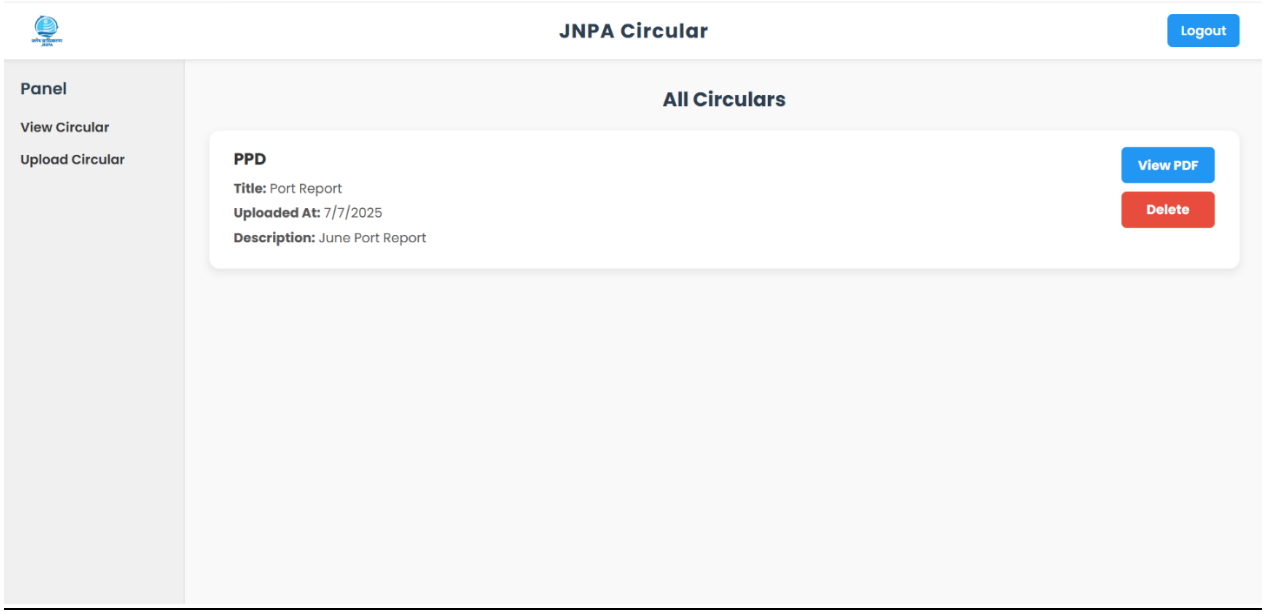


Fig.5 - Department View Circular

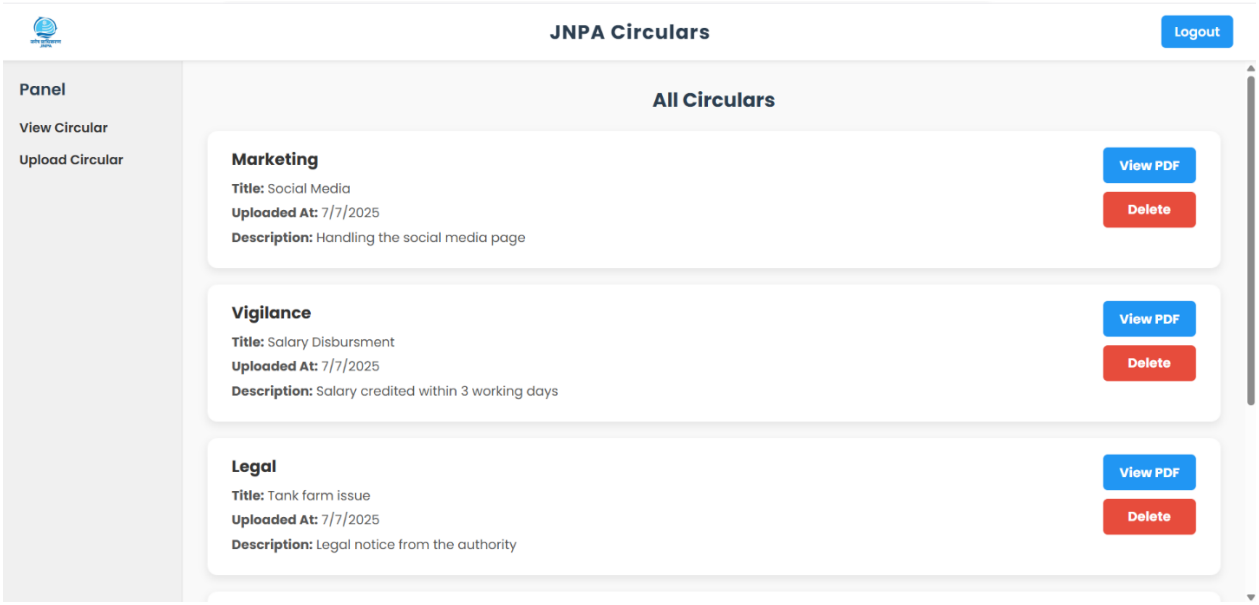


Fig.6 – Admin View Circular

Conclusion : The Circular Management System developed provides a streamlined, secure, and user-friendly solution for uploading, viewing, and managing departmental circulars within an organization. By eliminating manual processes and incorporating features such as role-based access, PDF uploads, and real-time updates, the system enhances administrative efficiency and transparency. The successful integration of a MySQL backend with a React-based frontend ensures robust performance and scalability. This project not only meets the current communication needs of departments but also lays a strong foundation for future enhancements like notification alerts, analytics, and mobile access

Conclusion

The internship experience at Jawaharlal Nehru Port Authority proved to be both enriching and insightful. It offered me practical exposure to enterprise IT operations, cybersecurity practices, and the use of professional tools like Oracle Financials and SAP. Through hands-on tasks and structured training sessions, I developed a clear understanding of system maintenance, network infrastructure, and secure information management within an organizational setting.

Working on the live project — the Circular Management System — enabled me to apply my knowledge of full-stack web development in a real-world context. From backend integration and database handling to frontend validation and role-based access, the project helped me gain valuable insights into designing scalable and efficient web applications.

In addition to enhancing my technical skills, the internship also improved my communication, teamwork, and time management abilities. It has equipped me with the confidence and experience required to work in a professional IT environment and laid a strong foundation for my future career in computer science.